

Sai Nitish Reddy Busireddy

Sr Data Engineer GenAI

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Professional Summary:

- Accumulated over 5 years of professional expertise in the IT sector, specializing in **Senior Data Engineer/Analyst roles**. Skilled in executing comprehensive end-to-end data engineering processes, including **Data Extraction, Transformation, and Ingestion** into structured data tables.
- Experienced in working with cloud platforms like Amazon Web Services, Azure, and Databricks, integrating with **Databricks** on both **Azure, GCP** and **AWS**.
- Strong background in migrating databases to Snowflake, with a deep understanding of **Snowflake** Database schema and table architecture.
- Skilled in installing, configuring, managing, supporting, and monitoring **Hadoop clusters** using distributions like **Apache Hadoop**, Cloudera Hortonworks, and cloud services such as **AWS, Azure** and **GCP**.
- Expertise in building **Azure** native enterprise applications, migrating applications to the Azure environment, and implementing **Data lakes** and **Business Intelligence** tools in Azure.
- Executed a data warehouse solution, involving **ETL** processes, on-premises to **cloud migration**, and deploying batch and streaming data pipelines in cloud environments.
- Utilized **Spark Data Frames, Spark SQL**, and **Spark's RDD API** for diverse data transformations and dataset construction.
- Proficient in integrating various data sources (**Oracle SE2, SQL Server, flat files, unstructured files**) within a data warehouse environment.
- Practical exposure to **NoSQL databases** like **HBase, Azure Cosmos DB, MongoDB**, and **Cassandra, Aws RedShift** understanding their functionality and implementation.
- Leveraged various features within **Teradata**, including **BTEQ, Fast Load, Multi Load, SQL Assistant, and DDL/DML commands, demonstrating** expertise in Teradata concepts.
- Proficiency in developing real-time data streaming solutions with **Apache Spark, Spark Streaming, Apache Storm, Kafka**, and **Flume**.
- Creation of **Power BI & Tableau** reports and impactful dashboards by collecting and translating end-user requirements.
- Collaboration with application architects to transition from **Infrastructure as a Service (IaaS)** to **Platform as a Service (PaaS) models**. Deployment of **Azure Resource Manager JSON Templates** using **PowerShell**.
- Solid experience in **Agile/Scrum methodologies**, actively participating in Scrum calls for project analysis and development discussions.
- Completed a job simulation involving AI-powered data analytics and strategy development for the Financial Services team at Tata iQ.
- Conducted exploratory data analysis (EDA) using GenAI tools to assess data quality, identify risk indicators, and structure insights for predictive modeling.
- Proposed and justified an initial no-code predictive modeling framework to assess customer delinquency risk, leveraging GenAI for structured model logic and evaluation criteria.
- Designed an AI-driven collections strategy leveraging agentic AI and automation, incorporating ethical AI principles, regulatory compliance, and scalable implementation frameworks.

Technical Skills:

Programming Languages	Python, Scala, SQL, Java, Shell Scripting
Python Libraries	NumPy, Pandas, Scikit-Learn, SciPy, Seaborn, Matplotlib,
Web Technologies	HTML, CSS, JavaScript, PHP
Hadoop Technologies	Hive, Spark, HDFS, Map Reduce, YARN, Kafka, Pig, Hive, Sqoop, HBase, Oozie, Zookeeper, Cloudera Manager and Horton Works
Cloud Platforms	Azure Cloud, AWS Cloud architecture, AWS Big Data computing, Microsoft Azure, GCP
Databases	Oracle, DB2, MySQL, Sybase, PostgreSQL, MS SQL Server, ETL Pipelines
AWS	EC2, S3, Glacier, Redshift, RDS, EMR, AWS Lambda, AWS Glue, CloudWatch, Recognition, Kinesis, CloudFront, DynamoDB, Code Pipeline, EKS, Athena, Quick Sight, IAM
Azure	Azure Storage, SQL Server, Data Factory, Databricks, Cosmos DB, Data Lake Storage, HDInsight, Stream Analytics, Synapse Analytics.
GCP	Cloud Storage, BigQuery, Cloud Dataflow, Cloud Pub/Sub, Cloud Dataproc, Composer, Cloud SQL, Google Cloud Datastore.

NoSQL Databases	MongoDB, Cassandra, HBase
Visualization Tools	Tableau, Power BI, Advanced Excel
Version Control Systems	Git, GitHub, CVS, SVN
IDEs	PyCharm, IntelliJ IDEA, Jupyter Notebooks, Eclipse, Visual Studio
Operating Systems	Unix, Linux, Windows
Methodologies	Agile/Scrum, Waterfall
LLMs/GenAI	OpenAI, LangChain, LlamaIndex, Hugging Face, Prompt Engineering, RAG
Vector DBs	FAISS, Pinecone, Weaviate
Security/Auth	MSAL, Azure AD, Microsoft Graph API
Evaluation	PromptEval, LangSmith, Trulens

Professional Experience:

Client – AAFES, Dallas, TX.

Dec 2022 – May 2025

Role – Data Engineer Gen AI

Responsibilities:

- Designed and implemented scalable **ETL pipelines** using **AWS Glue, PySpark, and Azure Data Factory**, ensuring efficient **data extraction, transformation, and loading (ETL)** across multiple environments.
- Developed and deployed **LLM-powered data applications** using OpenAI APIs and LangChain for automating ETL validation and metadata documentation tasks.
- Good understanding and experience of Gen AI concepts like Agentic Frameworks, RAG, LLM, Chunking, and Hyper-parameters.
- Prototyped knowledge retrieval bots using **retrieval-augmented generation (RAG)** techniques with **Pinecone** and **FAISS**, enabling dynamic document querying and summarization.
- Integrated **Microsoft Azure AD and MSAL** for secure access control and single sign-on across AI applications interacting with Microsoft 365 data via **Microsoft Graph API**.
- Experimented with **hybrid search strategies** combining semantic search, keyword filtering, and metadata-driven retrieval for improved GenAI app performance.
- Fine-tuned system prompts for domain-specific applications in **healthcare and contracting**, improving model precision and alignment with regulatory language.
- Developed and optimized **data warehousing** solutions using **Snowflake, Redshift, and S3**, improving query performance and data availability for business intelligence teams.
- Utilized **Spark Scala, Python, and NodeJS** for building high-performance data transformation workflows.
- Integrated AWS services such as **Kinesis, FireHose, IAM, and Lambda** to process and stream large volumes of structured and semi-structured data in real-time.
- Worked extensively with **non-relational databases**, including **MongoDB, HBase, and Cassandra**, implementing efficient data storage and retrieval mechanisms.
- Engineered **object storage** and **document store** solutions, leveraging **AWS S3, DynamoDB, and Azure Blob Storage** for scalable, cost-effective data archiving and access.
- Designed and maintained **key-value store, graph databases, and column-family databases**, optimizing performance for various data-driven applications.
- Designed and implemented a scalable ETL pipeline using PySpark on Azure Databricks, leveraging expertise in data modeling solutions and tools.
- Utilized Spark Scala scripts and User-Defined Functions (UDFs) to read data efficiently from Azure Blob storage in Azure Databricks, demonstrating in-depth knowledge of data modeling techniques.
- Designed and Implemented CI & CD Pipelines using Git, Jenkins, Maven, ECR and EKS from scratch achieving the end-to-end automation from code check into deploying docker images to EKS clusters.
- Implemented end-to-end data extraction, transformation, and loading (ETL) processes using Azure Data Factory (ADF) and Azure HDInsight, showcasing strong problem-solving skills in a fast-paced environment.
- Developed Python-based Spark applications for data processing, handled data from various RDBMS and streaming sources, and utilized Databricks notebooks for interactive analysis using Spark APIs.
- Implemented complex data transformation logic using Snowflake SQL and Python UDFs (User-Defined Functions) to cleanse, enrich, and harmonize incoming data from S3 sources. Configured and managed Azure Data Lake (ADLS) and Data Lake Analytics, integrating with other Azure services, and prepared scripts to automate ingestion processes from various sources such as API, AWS S3, Teradata, and Snowflake.
- Expertise in handling DB2 UDB security, establishing replication between servers, Capacity Planning, Backup and Restore, shipping archival logs, Corruption and Recovery, Managing login/permissions.
- Employed Azure DevOps and Jenkins pipelines for building and deploying different resources (code and infrastructure) in Azure, reflecting experience with agile software development methodologies.

- Designed and optimized Snowflake data warehouse schemas to support efficient querying and analytics across diverse datasets ingested from S3.
- Developed metrics dashboards and advanced filters in JIRA to provide performance metrics and status reports to end-users and business leadership.
- Moved data from Azure Blob storage to Azure Data Lake using Azure Data Factory (ADF) and performed ETL using PySpark on Azure Databricks, highlighting a comprehensive skill set in data engineering.

Environment: Linux, AWS EC2, S3, Python 3.6, R, pandas, SAP CRM, RAG, SRM, NLP, LLMs, Lang Graph, Crew AI, Vector DB Genism, SQL, Tableau, SQL, Pandas, Numpy, Sas, Iaas, Hadoop, Airflow, Docker, GCP Data Flow, Azure Data Factory, Snowflake, Terraform, Flink, Hadoop, SAS, Delta lake, PySpark, Kafta, HBase, Azure Cosmos DB, MongoDB, and Cassandra, Aws RedShift.

Client – Fidelity Investments Company,

May 2020– Aug 2021

Role: Data Engineer GCP, Azure, AWS(Contract)

Responsibilities:

- Designed and implemented a scalable ETL pipeline using PySpark on Azure Databricks, leveraging expertise in data modeling solutions and tools.
- Migrate Matillion pipelines and Looker reports from Amazon Redshift to Snowflake data warehouse.
- Utilized Spark Scala scripts and User-Defined Functions (UDFs) to read data efficiently from Azure Blob storage in Azure Databricks, demonstrating in-depth knowledge of data modeling techniques.
- Designed and Implemented CI & CD Pipelines using Git, Jenkins, Maven, ECR and EKS from scratch achieving the end-to-end automation from code check into deploying docker images to EKS clusters.
- Utilized Matillion ETL solution to develop pipelines that extract and transform data from multiple sources and load it to Snowflake.
- Migrated large-scale enterprise data to AWS S3 and Redshift, ensuring high-performance data processing.
- Designed and developed data ingestion pipelines using Spark, Databricks, and Hive.
- Automated data transformations using Python and PySpark for business intelligence applications.
- Worked extensively with AWS Lambda and API Gateway for serverless data orchestration.
- Implemented end-to-end data extraction, transformation, and loading (ETL) processes using Azure Data Factory (ADF) and Azure HDInsight, showcasing strong problem-solving skills in a fast-paced environment.
- Developed Python-based Spark applications for data processing, handled data from various RDBMS and streaming sources, and utilized Databricks notebooks for interactive analysis using Spark APIs.
- Implemented complex data transformation logic using Snowflake SQL and Python UDFs (User-Defined Functions) to cleanse, enrich, and harmonize incoming data from S3 sources. Configured and managed Azure Data Lake (ADLS) and Data Lake Analytics, integrating with other Azure services, and prepared scripts to automate ingestion processes from various sources such as API, AWS S3, Teradata, and Snowflake.
- Expertise in handling DB2 UDB security, establishing replication between servers, Capacity Planning, Backup and Restore, shipping archival logs, Corruption and Recovery, Managing login/permissions.
- Employed Azure DevOps and Jenkins pipelines for building and deploying different resources (code and infrastructure) in Azure, reflecting experience with agile software development methodologies.
- Designed and optimized Snowflake data warehouse schemas to support efficient querying and analytics across diverse datasets ingested from S3.
- Extensively used advance chart visualizations in Tableau like dual axis, Box Plots, Bullet Graphs, Tree Maps, Bubble charts, Waterfall charts, funnel charts etc.
- Moved data from Azure Blob storage to Azure Data Lake using Azure Data Factory (ADF) and performed ETL using PySpark on Azure Databricks, highlighting a comprehensive skill set in data engineering.

Environment: AWS Cloud Services, Apache Spark, Spark-SQL, Unix, Kafka, Scala, SQL Server, Azure Data center, AWS S3, Aws Glue, Aws Lambda, PostgreSQL, Docker, GCP Data Flow, Azure Data Factory, Snowflake, Terraform, Flink, Hadoop, SAS, Delta lake, PySpark.

Certifications and Licenses:

Tata Group Data Analytics Job Simulation on Forage - June 2025

Education:

Bachelor --- Sathyabama Institute of Science and Technology, India, March 2020.

Masters --- University of North Texas, TX, USA, Dec 2022.